

Name: Recitation 09/01/2023

(Suha's answer! Open to any comments! Let me know.)

1. **Consideration sets:** Suppose an individual has alternatives given by $X = \{a, b, c, d, e\}$ and a domain of $\mathcal{D} = \{\{a, b\}, \{a, b, c\}, \{a, b, c, d\}, \{a, b, c, d, e\}\}$. Suppose an individual only considers a subset of alternatives from each menu. The subset of alternatives considered from each menu is given by the following mapping $\Gamma : \mathcal{D} \rightarrow X$. We have consideration sets given by

M	$\Gamma(M)$
$\{a, b\}$	$\{a\}$
$\{a, b, c\}$	$\{b, c\}$
$\{a, b, c, d\}$	$\{c, d\}$
$\{a, b, c, d, e\}$	$\{a, b, c, d, e\}$

and transitive and complete preferences given by the order $e \succ d \succ c \succ b \succ a$. Suppose one individual chooses according to $C_{\succ}(M) = \{x \mid x \succ y \ \forall y \in M \setminus \{x\}\}$ and $C_{\Gamma, \succ} = \{x \mid x \succ y \ \forall y \in \Gamma(M) \setminus \{x\}\}$.

- (a) Compute the choices from each menu in the domain according to $C_{\succ}$. *"I think this 'a' comes from $T(M)$ (not M) as it is excluded from $T(M)$ in the first place when considering 'y'."*

M	$C_{\succ}(M)$
$\{a, b\}$	b
$\{a, b, c\}$	c
$\{a, b, c, d\}$	d
$\{a, b, c, d, e\}$	e

- (b) Compute all the choices from each menu in the domain according to $C_{\Gamma, \succ}$.

M	$C_{\Gamma, \succ}$
$\{a, b\}$	a
$\{a, b, c\}$	c
$\{a, b, c, d\}$	d
$\{a, b, c, d, e\}$	e

- (c) Suppose that you have observed the choices from the two choice functions and generate a revealed preference where $x R y$ when for some menu M we have $x \in C(M)$ and $y \in M$. How do the two revealed preference relations compare?

For $C_{\succ}(M)$:

R	a	b	c	d	e
a	?	b	c	d	e
b		b	c	d	e
c			c	d	e
d				d	e
e					e

For $C_{\Gamma, \succ}(M)$:

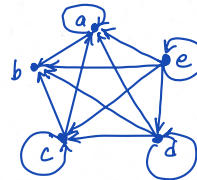
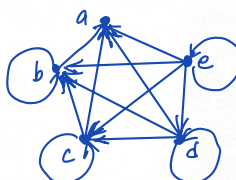
R	a	b	c	d	e
a	a	a	c	d	e
b	a	?	c	d	e
c			c	d	e
d				d	e
e					e

Both are revealed incomplete but transitive.

To see transitivity:

$$(x \rightarrow y \Leftrightarrow x > y)$$

look for "triangles" 



(d) What happens when the above holds but $\Gamma(\{a, b, c, d, e\}) = \{b, c\}$

We can reveal from $C_P, >$ that:

R	a	b	c	d	e
a	a	a	c	d	?
b	a	?	c	d	?
c			c	d	?
d			c	d	?
e			c	d	?

If we define the indifference as $x \sim y \Leftrightarrow x R y \& y R x$, then $c \sim d$ may be revealed.

2. Convexity:

(a) Suppose the function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is concave. Prove that f is quasi-concave.

pf) Let $x, y \in \mathbb{R}^N$. WLOG, we may assume $f(x) \leq f(y)$.

Let $\lambda \in [0, 1]$ be given.

(We shall show: $f(\lambda x + (1-\lambda)y) \leq f(y)$ by the def of quasi-concavity.)

Then

$$\begin{aligned} f(\lambda x + (1-\lambda)y) &\leq \lambda f(x) + (1-\lambda)f(y) \quad (\text{by the def of concavity}) \\ &\leq \lambda f(y) + (1-\lambda)f(y) \quad (\because f(x) \leq f(y)) \\ &= f(y) \end{aligned}$$

Thus, f is quasi-concave.

Graded!

(b) Suppose $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is concave and $g : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic increasing. Is $g \circ f$ concave? If so, prove it. If not, find a counter example and explain why.

False.

(Counter-example)

$$f(x) = \ln x^2$$

$$g(x) = e^x$$

$$\text{Then } g \circ f(x) = e^{\ln x^2} = x^2 : \text{ a convex function.}$$

$$\delta_C: C \rightarrow \mathbb{R} \quad \delta_C: \mathbb{R}^N \rightarrow \mathbb{R}$$

(c) Consider the function

$$\delta_C(y) = \begin{cases} 0 & \text{if } y \in C \\ -\infty & \text{otherwise} \end{cases}$$

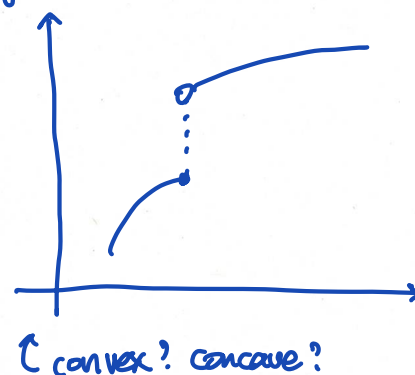
where $C \subset \mathbb{R}^N$. Can you come up with a condition on C so the function is convex? Prove C is convex for the case you study.

Prop If a function is concave/convex,
the function is continuous.

$\Leftrightarrow$ If Not continuous, not convex! $\rightarrow$ e.g.,

To have $\delta_C(y)$ continuous,

$C = \emptyset$ or $\mathbb{R}^N$.



Suha's Note

2. **Stochastic choice:** Consider $X = \{a, b, c, d\}$ and the domain of menus is given by $\mathcal{D} = \{\{a, b\}, \{c, d\}, \{a, b, c\}, \{b, c, d\}, \{a, b, c, d\}\}$. Moreover, suppose we have a distribution over strict preference orders given by

$p(\succ)$	$\succ$
$1/3$	$a \succ_1 b \succ_1 c \succ_1 d$
$1/4$	$d \succ_2 b \succ_2 c \succ_2 a$
$1/4$	$c \succ_3 a \succ_3 d \succ_3 b$
$1/6$	$a \succ_4 d \succ_4 b \succ_4 c$

where stochastic choice from menu M is given by

$$p(x \mid M) = \sum_{i=1}^4 p(\succ_i) 1\{x \succ_i y \quad \forall y \in M \setminus \{x\}\}.$$

- (a) (20 points) Compute stochastic choice probabilities for every alternative and menu in the domain

$$p(a \mid \{a, b\}) = \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{3}{4}$$

$\vdots$

(See $\otimes$ at the bottom of the questions.)

- (b) (10 points) Can you figure out any properties on choice probabilities $p(x \mid M)$ and $p(x \mid N)$ when $N \subseteq M$?

$$\text{If } N \subseteq M, \quad p(x \mid N) \geq p(x \mid M).$$

Yes. Hold for other prob. dist.

(c) (10 points) Would the property you found in (b) hold for other probability distributions over preferences? Describe why or why not you think this would hold.

- Among the menu, Consider only the case of $N \subseteq M$. WLOG, we may consider when $N = \{x, y\}$, $M = \{x, y, z\}$, where $\{x, y, z\} = \{a, b, c\}$.
- The ordering b/t x, y includes the following cases:
 $x > y \iff \begin{cases} \text{i) } x > y > z : p_1 \\ \text{ii) } x > z > y : p_2 \\ \text{iii) } z > x > y : p_3 \end{cases}$
 - If we have the probabilities attached to each orderings i) ~ (iii), the overall probabilities of $x > y$ should be $p_1 + p_2 + p_3$, which is greater or equal to each p_1, p_2 , or p_3 .

(d) (10 points) Is it possible to find a distribution over different preference orderings that gives the same distribution? Sketch why you think this is true or false.

The structure of the menu enables this.

$$\mathcal{D} = \{\{a, b\}, \{c, d\}, \{a, b, c\}, \{b, c, d\}, \{a, b, c, d\}\}$$

- We do not have menu $\{b, c\}$ in the domain, so we do not consider the distribution over choice b/t b & c .

To have different preference orderings to be taken into account while preserving the same choice probabilities for each menu, we need to consider the following:

- When expanding $\{a, b\}$ to $\{a, b, c\}$,
 adding "c" must not affect the most preferred one of a & b .
 When expanding $\{c, d\}$ to $\{b, c, d\}$,
 adding "b" must not affect the most preferred one of c & d .

($\therefore$ same holds for all the set inclusions)

- We may change the preference orderings (with the same weights) as long as the orderings satisfies the above qualification.
- For example, we may consider the following distributions
 $p'(>)$ or $p''(>)$ over different preference orderings:
 ( : the parts that are different from the original)

Original distribution 2

$p(>)$	$>$
$\frac{1}{3} = p_1$	$a >_1 b >_1 c >_1 d$
$\frac{1}{4} = p_2$	$d >_2 b >_2 c >_2 a$
$\frac{1}{6} = p_3$	$c >_3 a >_3 d >_3 b$
$\frac{1}{6} = p_4$	$a >_4 d >_4 b >_4 c$

$$p_1 + p_2 + p_3 + p_4 = 1$$

$p'(>)$	$>$
p_1	$a >_1 b >_1 c >_1 d$
p_2	$d >_2 c >_2 b >_2 a$
p_3	$c >_3 a >_3 b >_3 d$
p_4	$a >_4 d >_4 c >_4 b$

$p''(>)$	$>$
p_1	$a >_1 b >_1 c >_1 d$
p_2	$d >_2 c >_2 b >_2 a$
p_3	$c >_3 d >_3 a >_3 b$
p_4	$a >_4 d >_4 c >_4 b$

$$\Rightarrow p(a | \{a, b\}) = p_1 + p_3 + p_4$$

$$p(b | \{a, b\}) = p_2$$

$$p(c | \{c, d\}) = p_1 + p_3$$

$$p(d | \{c, d\}) = p_2 + p_4$$

$$p(a | \{a, b, c\}) = p_1 + p_4$$

$$p(b | \{a, b, c\}) = p_2$$

$$p(c | \{a, b, c\}) = p_3$$

$$p(b | \{b, c, d\}) = p_1$$

$$p(c | \{b, c, d\}) = p_3$$

$$p(d | \{b, c, d\}) = p_2 + p_4$$

$$p(a | \{a, b, c, d\}) = p_1 + p_4$$

$$p(b | \{a, b, c, d\}) = 0$$

$$p(c | \{a, b, c, d\}) = p_3$$

$$p(d | \{a, b, c, d\}) = p_2$$

⊗

• The distributions $p'(>)$ or $p''(>)$ give the same choice probabilities.

Original:

$\frac{1}{3}$ ①

$a > b > c > d$

①

$a > b > d > c$

$\frac{1}{6}$ ④

$a > d > b > c$

$a > d > c > b$

④

$a > c > b > d$

$a > c > d > b$

$b > a > c > d$

$b > a > d > c$

$b > c > a > d$

$b > c > d > a$

$b > d > a > c$

$b > d > c > a$

$\frac{1}{6}$ ③

$c > a > b > d$

$c > a > d > b$

$c > b > a > d$

$c > b > d > a$

$c > d > a > b$

$c > d > b > a$

③

$d > a > b > c$

$d > a > c > b$

$d > b > a > c$

$d > b > c > a$

$d > c > a > b$

$d > c > b > a$

②

$\frac{1}{4}$ ②